Vinoothna Nadikatla

AI Engineer | Machine Learning Developer | Data Scientist | Data Analyst

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Professional Summary

Applied AI M.Tech candidate with strong hands-on experience in machine learning, deep learning, and Generative AI systems. Demonstrated ability to design and deliver explainable, production-ready ML solutions across predictive maintenance, cybersecurity, and multimodal AI domains. Experienced in building end-to-end ML pipelines, including data preprocessing, model training, evaluation, deployment, and explainability.

Education

Visvesvaraya National Institute of Technology (VNIT) - Applied AI — July 2024 – May 2026

- GPA: 8.27/10.0

Rajiv Gandhi University of Knowledge Technologies - Mechanical Engineering — July 2020 – May 2024

- GPA: 7.3/10.0

Technologies

Languages: Python, Java, Oops, SQL
AI/ML: Scikit-learn, XGBoost, TensorFlow, PySpark, Explainable AI (XAI), SHAP, Computer Vision, MLOps, MLflow, Big Data, NLP, Deep Learning, Gen AI, Prompt Engineering, LLM, RAG.
Tools/Platforms: MySQL, Power BI, Streamlit, APIs, Git/GitHub, Azure, Docker, DVC, CI/CD, Kubernetes, Data Analysis, Cybersecurity.

Publications

"Explainable Hybrid Gradient Boosting–Deep Learning Framework for Bearing Fault Classification" — publication in PCEMS International Conference 2025; published in Springer LNNS proceedings and indexed in IEEE Xplore, highlighting contributions to applied AI and engineering research. — Nov 2025

Projects

Agentic Multimodal AI Assistant

- Built an agentic AI assistant integrating LLM reasoning, speech (STT/TTS), and computer vision. Designed multi-step workflows using LangGraph for decision-making and task orchestration. Enabled real-time interaction and contextual reasoning.
- **Tools Used:** Python, Groq, Lang Graph, CV, STT, TTS

VN AI Safety Monitor — Agentic Multimodal GenAI System

- Built an agentic multimodal AI safety monitor using LangGraph, Groq LLMs, YOLOv8, and FastAPI with real-time PPE detection, multilingual voice interaction, and MJPEG streaming on commodity hardware.
- **Tools Used:** Python, LangGraph, Groq API, Llama-3.3-70b, Llama-4-Scout-17b, Whisper, YOLOv8n, OpenCV, EasyOCR, MJPEG, gTTS, Git

Explainable Hybrid Gradient Boosting–Deep Learning Framework for Bearing Fault Classification

- Developed a 98.34%-accurate Gradient Boosting predictive maintenance model with hybrid features and SHAP explainability, delivering a lightweight real-time industrial fault diagnosis system.
- **Tools Used:** Python, Scikit-learn, TensorFlow, NumPy, SciPy, SHAP, Wavelet Packet Decomposition.

Cybersecurity Incident Classification- Microsoft

- Developed an ML model using the Microsoft GUIDE dataset to classify cybersecurity incidents (True Positive, False Positive, Benign). Implemented feature engineering, XGBoost optimization, and SMOTE balancing, achieving a high detection accuracy. Automated Security Operations Centre (SOC) incident triage, reducing response time by 30%.
- **Tools Used:** Python, Scikit-learn, matplotlib, seaborn, XGBoost, imbalanced-learn.

YouTube Data Harvesting and Warehousing

- Extracted & analyzed YouTube channel data from 15+ channels using APIs & SQL. Automated structured storage, enabling 20% times faster data reporting.
- **Tools Used:** Python, SQL, APIs, Streamlit.

Used Car Price Prediction- Car Dekho

- Built a machine learning model & Streamlit app for predicting used car prices with 90% accuracy. Processed 10,000+ data entries, performing Data Cleaning, EDA, and Feature Engineering.
- **Tools Used:** Python, Scikit-learn, matplotlib, seaborn, Streamlit.

Certifications

- Prompt Engineering for ChatGPT – *Coursera*,
- Generative AI – *GUVI (IIT Madras)*,
- Principles of Generative AI – Infosys,
- Robotics Internship - *MVARO (Verzeo)*
- 3Dprinting – *Garuda3D*